

SkillSwap: Peer Learning

1. Problem Definition:

In today's digital era, individuals continuously seek opportunities to learn new skills and enhance their professional growth. However, many people are unable to access quality learning resources due to the high cost of online courses. In contrast, others possess valuable skills but have limited opportunities to share their knowledge in a meaningful way.

Current learning platforms primarily support one-way knowledge delivery and monetary transactions, leaving little scope for collaborative, peer-to-peer skill exchange. As a result, learners and skill providers often struggle to connect, exchange expertise, and build long-term learning relationships.

SkillSwap is proposed as a centralized, barter-based skill-exchange platform that connects users based on complementary skills and learning interests. By enabling users to teach what they know in exchange for learning what they need, the platform promotes accessible learning, knowledge sharing, and a collaborative community through a secure and transparent exchange system.

2. Features:

- Secure User Registration & Authentication
- User Profile & Skill Management
- Intelligent Skill Matching
- Skill Swap Request & Management
- Admin Dashboard for Platform Management

3. Objectives :

- To provide a platform for peer-to-peer skill exchange.
- To connect users based on their skills and learning interests.
- To allow users to offer and learn skills without relying primarily on monetary transactions.
- To facilitate skill-swap requests between users.
- To provide a secure and transparent environment for knowledge sharing.

4. Technologies Used:

Topics	We Used
Frontend	React
Backend	Java, Spring Boot framework
Security	Spring Security, JWT Authentication
Database	Neon PostgreSQL (Cloud Database)
ORM	Spring Data JPA (Hibernate)
APIs	REST API
Build Tool	Maven
Version Control	Git, GitHub
Documentation	Swagger / OpenAPI

5. Stakeholders:

- Learners
- Skill Providers
- Platform Administrator

6. Expected Outcomes:

- Promote affordable and accessible skill development.
- Enable effective peer-to-peer skill exchange.
- Improve collaboration through intelligent skill matching.
- Build a trusted learning community.
- Provide a secure and scalable platform for skill sharing.

7. Internal wise module Separation:

Internal-1: Core Modules	Internal-2: Advanced & Supporting Modules
Auth Module	Dashboard Module
Profile Module	Smart Match Module
Browse Skill Module	Session Module
Swap Request Module	Chat Module <i>(Optional)</i>
	Review Module <i>(Optional)</i>